S5 Appendix. Additional conceptual figures

S5.1. Biological motivation for a slow accessibility variable

The figure below situates M_b as a phenomenological compression of slow branch-local processes: spine-neck geometry, actin remodeling, dynamic microtubule entry, local translation, transport readiness, and mitochondrial support. None of these processes is required to map one-to-one onto M_b. Rather, they collectively motivate the class of slow branch-local access constraints that M_b represents.

The figure is provided for conceptual orientation. It does not represent a direct biological measurement, a validated model of any specific molecule, or a claim that cytoskeletal structures directly store memory content. The main article's claim rests on simulator discrimination, not on the biological sketch.

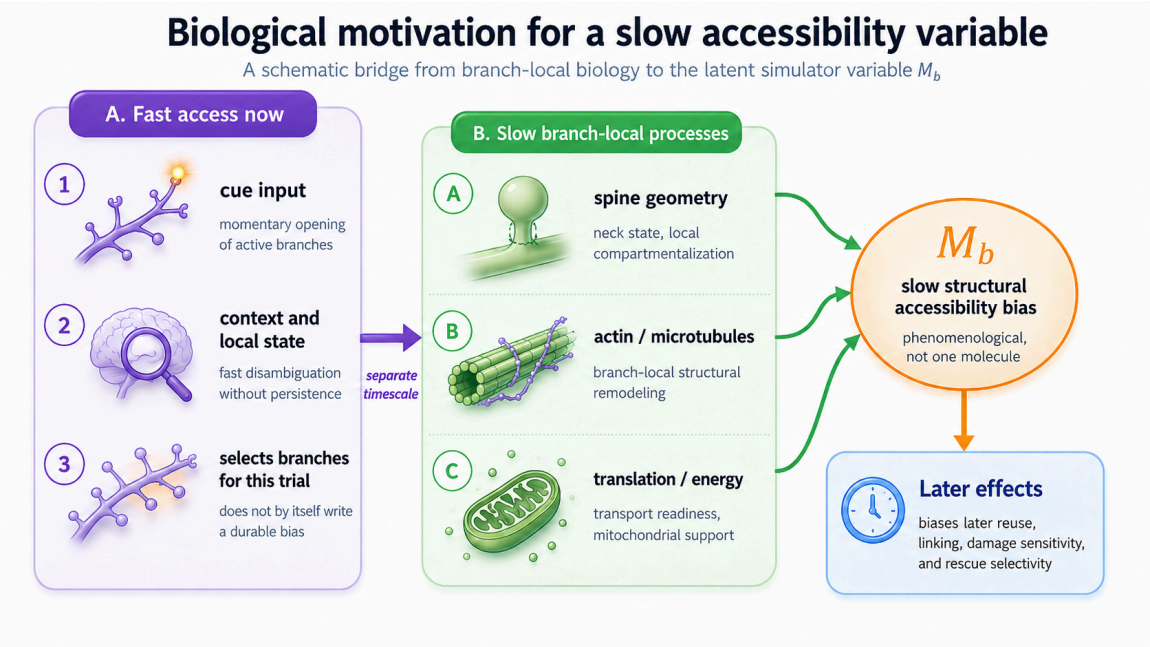


Figure S5.1: **Biological motivation for the slow accessibility variable M_b.** Fast branch opening is driven by cue input, contextual gating, and local dendritic/spine state (left side). Slow branch-local processes — spine geometry, actin dynamics, dynamic microtubule entry, local translation, transport readiness, and metabolic support — evolve on longer timescales and are compressed into the latent variable M_b (right side). M_b is a phenomenological variable: it is biologically motivated and operationally defined, but it should not be read as a direct molecule count or as proof of a cytoskeletal memory code. Candidate processes include spine-neck compartmentalization [1], [2], [3], actin-dependent spine remodeling [4], [5], dynamic microtubule entry into spines [6], [7], [8], [9], [10], branch-local translation [11], [12], [13], and mitochondrial/metabolic support [14], [15], [16].

S5.2. Structural impedance gating at the spine neck

Microtubule-associated remodeling is one illustrative candidate contributor to structural accessibility [6], [7], [8], [10], [17]. In the schematic below, a low-support structural state is depicted as higher effective impedance at the spine neck, limiting voltage spread from dendrite into spine [1], [2], [3], [18]. A high-support structural state is depicted as lower effective impedance, permitting signal propagation. This contrast illustrates how slow branch-local

structural changes could gate signal flow independently of fast synaptic input strength. The fast/slow access factorization in the model is motivated by this class of slow-timescale process; M_b compresses any combination of them.

This figure is illustrative. The model does not require MT dynamics specifically; M_b may represent any combination of slow branch-local processes.

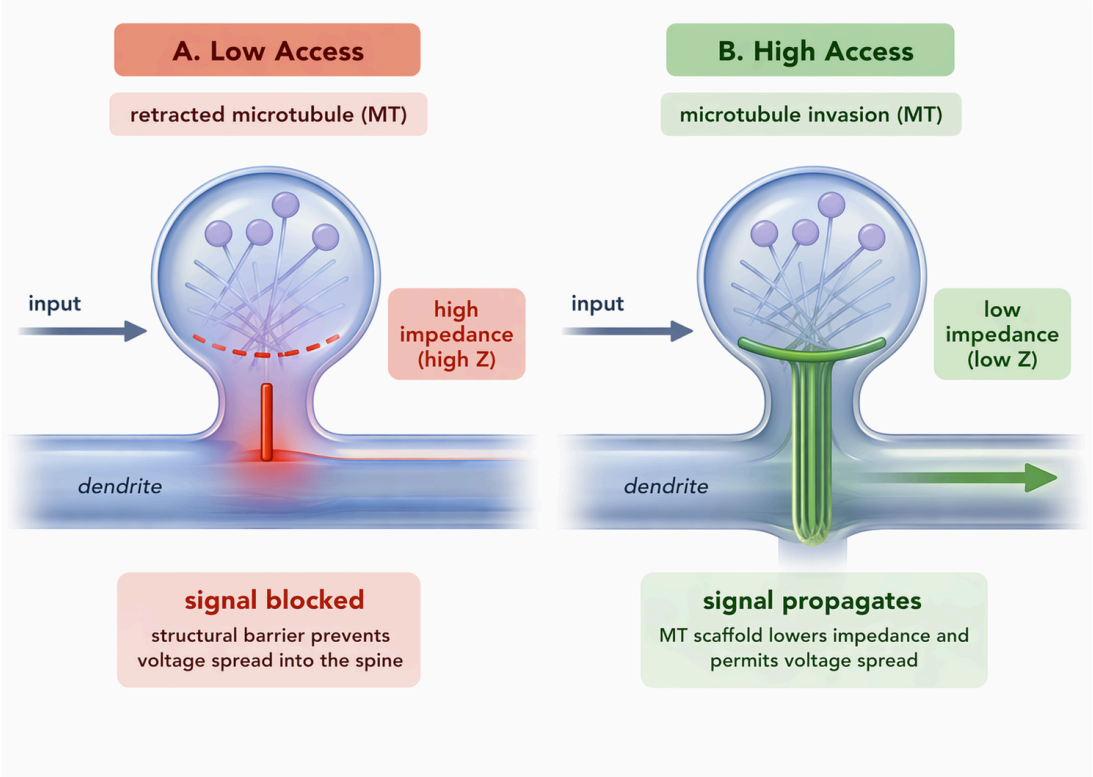


Figure S5.2: **Structural impedance gating at the spine neck (schematic). (A)** Low-support state: the spine-neck structural configuration is schematized as higher effective impedance, limiting voltage spread from dendrite into spine. **(B)** High-support state: a structurally permissive configuration is schematized as lower effective impedance, permitting signal propagation. Microtubule-associated remodeling is one illustrative candidate contributor; the schematic does not assert that MT dynamics are the unique or dominant mechanism. M_b compresses this and related slow-timescale branch-local processes into a single latent variable.

S5.3. Minimal biological formalization: variable structure and equations

This historical schematic illustrates a broader theoretical formulation, not the discrete updates executed in E017–E025. In particular, its x_b is an ungated integration state and its readout includes A_b explicitly. In the implemented simulator, x_b already includes access gating, so recall support does not multiply by A_b again. See S1, Mapping from the continuous family to the executable simulator, for the limitations of the correspondence and the exact discrete update equations.

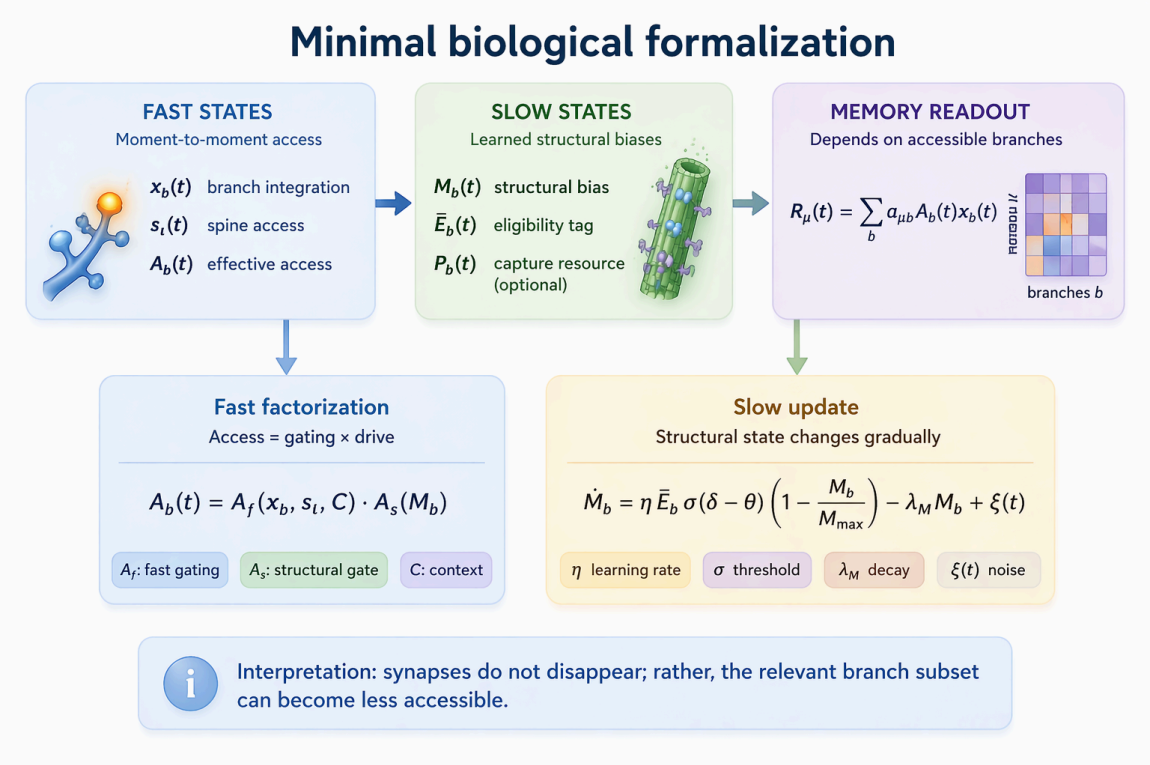


Figure S5.3: **Historical theoretical formalization: variable structure and update equations.** This diagram is not the executed E017–E025 readout; see the explanatory text above. Left: fast states (x_b branch integration, s_t spine access, A_b effective access). Centre: slow states (M_b structural bias, $\overline{E}_b$ eligibility tag, P_b capture resource). Right: memory readout $R_\mu = \sum_b a_{\mu b} A_b x_b$. Bottom left: fast factorization $A_b = A_{f(x_b, s_t, C)} \cdot A_{s(M_b)}$. Bottom right: slow update $\dot{M}_b = \eta \overline{E}_b \sigma(\delta - \theta)\left(1 - \frac{M_b}{M_{\max}}\right) - \lambda_M M_b + \xi(t)$. Interpretation note: synapses do not disappear; rather, the relevant branch subset can become less accessible.

S5.4. Memory linking and contextual retrieval via shared branches

This figure illustrates the key operative prediction: two episodes encoded close in time that share a dendritic branch segment can become linked, so that retrieving one helps recruit the other. An unrelated episode (Episode C) that does not share a branch remains independent. This is the core testbed for the slow-accessibility hypothesis and provides intuition for the simulator's two-trace overlap motif.

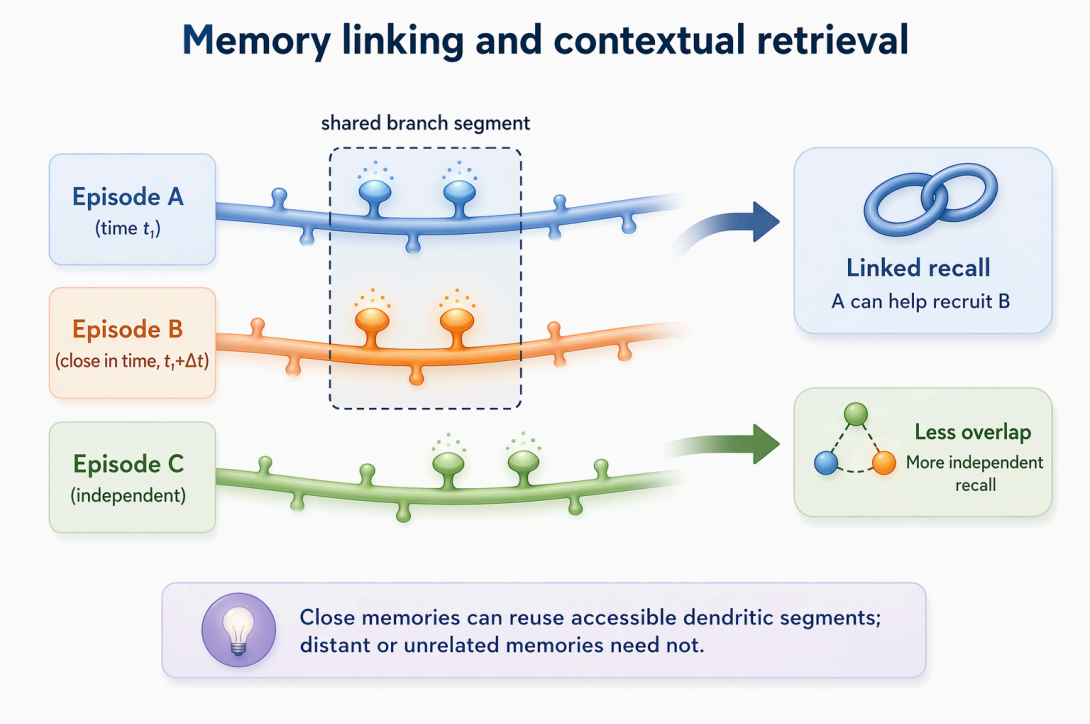


Figure S5.4: **Memory linking and contextual retrieval via shared dendritic branches.** Episodes A and B (encoded close in time, t_1 and $t_1 + \Delta t$) share a branch segment (dashed box). Replay-dependent slow structural writing raises M_b on the shared branch, increasing the linking score $L_{\mu\nu}$; retrieval of A can recruit B (linked recall, upper right). Episode C (independent, no shared branch) remains unlinked (lower right). This schematic motivates the two-trace overlap motif used throughout the simulator experiments and the generalized motif tests in E021–E021R. Memory linking via shared dendritic structure is supported by work on engram neurons, contextual linking, and hippocampal memory allocation [19], [20], [21], [22], [23], [24], [25].

S5.5. Evidence ladder and claim boundary

The figure separates historical internal mechanism diagnostics and boundary analyses from the primary E025 common-probe comparison. The equal-budget, locked-holdout comparison does not demonstrate a unique slow-writing advantage on its defined task. The evidence is limited to the tested implementations, adapters, parameter ranges, and generated topologies; S3 open-data bridges do not validate the latent state.

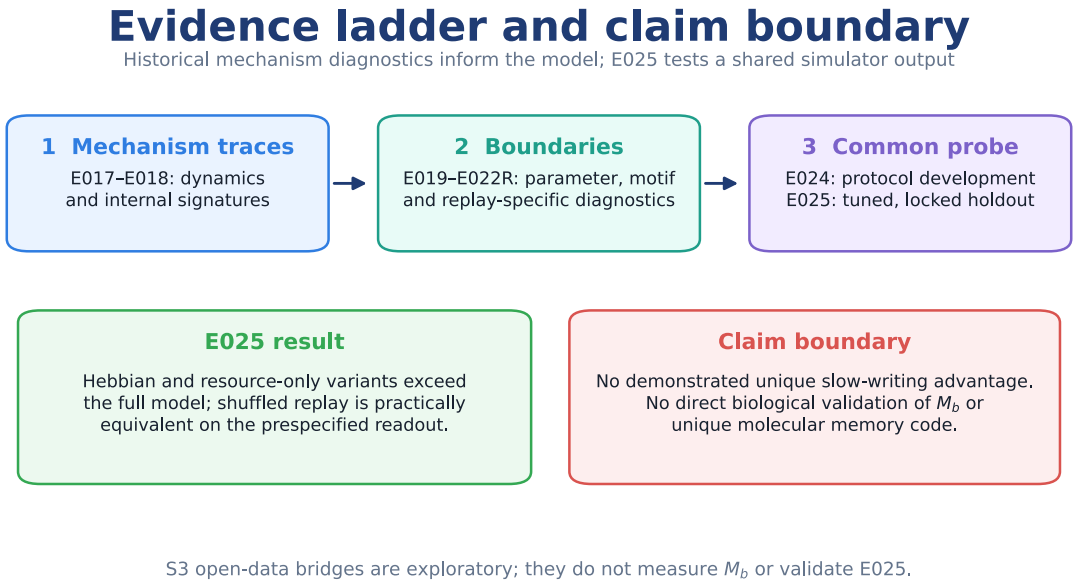


Figure S5.5: **Evidence ladder and claim boundary.** E017–E022R characterize simulator dynamics, internal mechanism signatures, and parameter and motif boundaries. E024 develops a common probe; E025 compares tuned model families on a locked holdout. Hebbian and resource-only variants exceeded the full model on the primary readout, while shuffled replay was practically equivalent under the registered decision rule. No biological measurement of M_b or unique molecular memory code is established.

S5.6. Overlap damage and targeted rescue logic

This figure summarizes the mechanistic logic behind SIG-D and SIG-E. Two traces share an overlap branch that is selectively strengthened by replay-linked slow writing. Focal damage to the shared branch disproportionately reduces linking relative to single-trace recall (SIG-D). Overlap-targeted rescue restores linking more effectively than generic rescue applied to non-overlap branches (SIG-E).

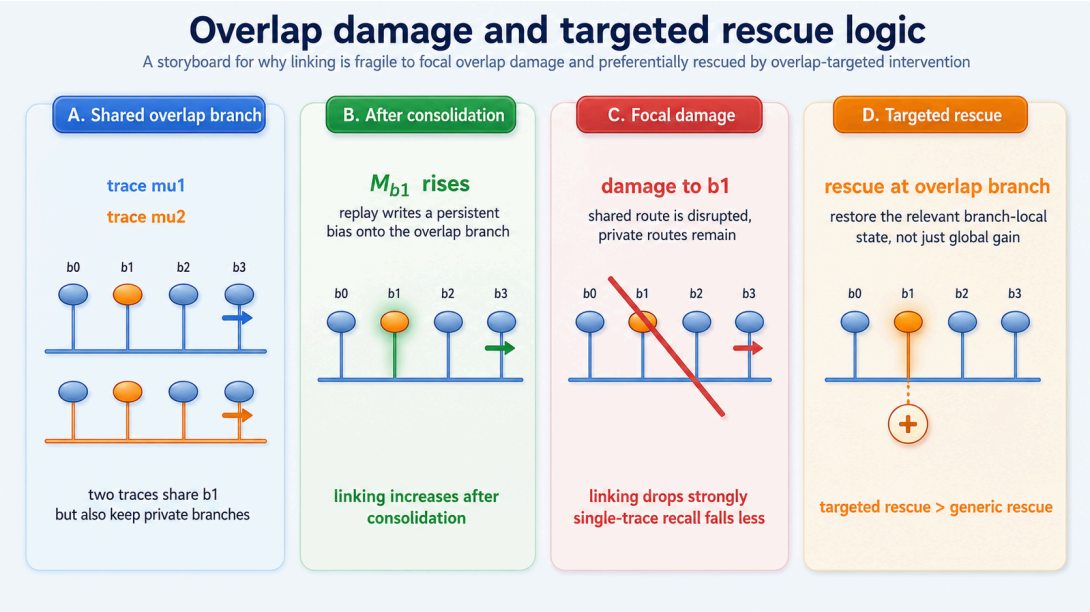


Figure S5.6: **Overlap damage and targeted rescue logic.** Two traces (A, B) share an overlap branch whose slow structural accessibility M_b has been raised by replay-dependent consolidation. Focal damage to the overlap branch selectively reduces the linking score L_{AB} while largely preserving single-trace recall support R_A and R_B (SIG-D). Targeted rescue — which restores eligibility on the overlap branch — recovers linking more than non-targeted generic rescue (SIG-E). This is a historical mechanism-specific diagnostic; E025 uses a separate common-probe comparison and does not test this rescue protocol across all families.

S5.7. Comparator evaluation and model-discrimination logic

E025 compares specified model families using a shared probe and readout after equal-budget tuning. Historical E018/E022 mechanism signatures and proxy outcomes describe internal behavior under their original protocols; they are not the E025 common-readout test and cannot by themselves establish an output-level advantage for slow writing.

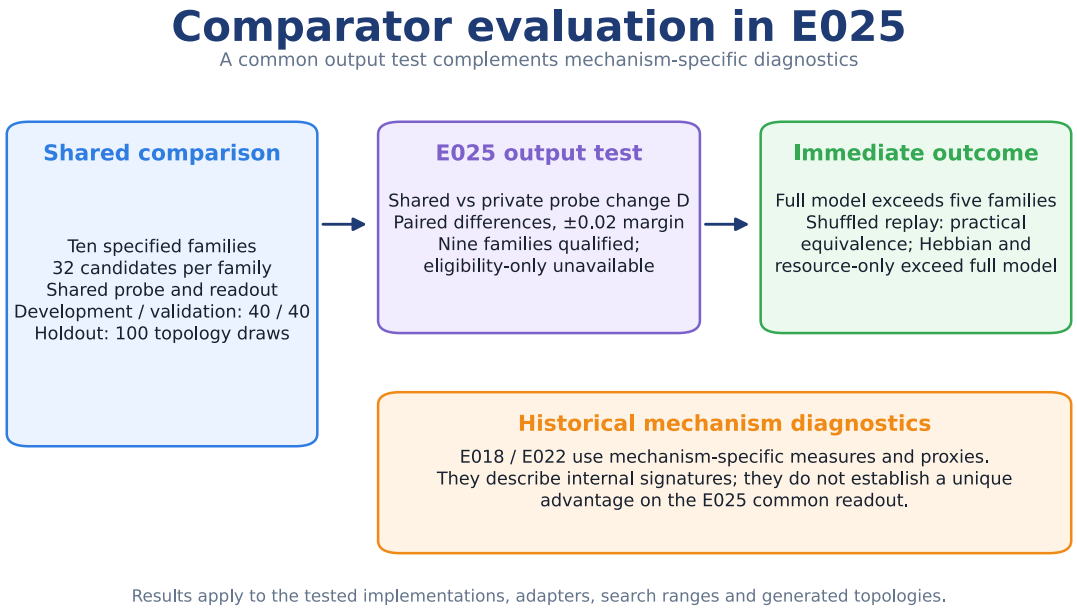


Figure S5.7: **Comparator evaluation and model-discrimination logic.** E025 used a shared probe, model-defined adapters, 32 candidate configurations per family, and independent development, validation, and holdout topology draws. On the immediate readout the full model exceeded five testable alternatives, was practically equivalent to shuffled replay, and was exceeded by Hebbian and resource-only variants. Eligibility-only had no qualifying candidate. These are outcomes for the specified implementations and the registered decision rule, conditional on the sign-flip assumption; S2 reports the post-hoc bootstrap sensitivity and further limits.

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